Homework Section 15.9

1. Find the Jacobian of the transformation: $x = u^2 - v^2$, $y = u^2 + v^2$.

2. Find the image of the set under the given transformation: S is the triangular region with vertices (0, 0), (1, 1), (0,1); $x = u^2$, $y = v$.

3. Use the given transformation to evaluate the integral:

   a) $\iint_R x^2 dA$, where R is the region bounded by the ellipse $9x^2 + 4y^2 = 36$; $x = 2u$, $y = 3v$.

   b) $\iint_R xy dA$, where R is the region in the first quadrant bounded by the lines $y = x$, $y = 3x$ and the hyperbolas $xy = 1$, $xy = 3$; $x = \frac{u}{v}$, $y = v$.

   c) $\iint_R (x^2 - xy + y^2) dA$, where R is the region bounded by the ellipse $x^2 - xy + y^2 = 2$; $x = \sqrt{2}u - \sqrt{\frac{2}{3}}v$ and $y = \sqrt{2}u + \sqrt{\frac{2}{3}}v$

4. Evaluate the integral by making an appropriate change of variables.

   a) $\iint_R \frac{x - 2y}{3x - y} dA$, where R is the parallelogram enclosed by the lines $x - 2y = 0$, $x - 2y = 4$, $3x - y = 1$, $3x - y = 8$.

   b) $\iint_R \cos\left(\frac{y - x}{y + x}\right) dA$, where R is the trapezoidal region with vertices (1, 0), (2,0), (0, 2), and (0, 1).